PRODUCT ARCHITECTURE • PREPSTACK

Five Connected Systems

A job-oriented architecture for turning preparation into demonstrable readiness

01	02	03	04	05
JOB INTELLIGENCE	PREPARATION	SKILL VERIFICATION	PROJECT ↔ JOB ALIGNMENT	INTERVIEW READINESS

The central idea

Five engines. One shared skill catalog. One connected journey from job requirements to verified ability.

PRODUCT DIRECTION & IMPLEMENTATION PRIORITIES

THE PRODUCT VISION

Organise PrepStack around five connected systems

PrepStack should not feel like five separate tools placed under one navigation bar. Each system should contribute information to the others, creating a continuous job-preparation workflow.

> **Target a role → understand the gaps → prepare → verify skills → build relevant project evidence → demonstrate readiness in interviews.**

The five systems below describe the product architecture and the responsibilities each part of PrepStack should own. They are connected through a shared Skill Catalog and a common record of learning, practice and evidence.

01 Job Intelligence Engine

The Role Analyzer evolves into the starting point of preparation.

- Resume vs. job matching
- Skill-gap identification
- Requirement prioritisation
- Target-role analysis

02 Preparation Engine

Connect learning resources to the actual skills and requirements identified for a target role.

- DSA preparation
- Core CS subjects
- Roadmaps
- Behavioural preparation
- My Plan

03 Skill Verification Engine

Test whether students understand and can apply what they have studied.

- Topic assessments
- Coding assessments
- Practical exercises
- Conceptual explanations
- Skill-level evidence

04 Project-to-Job Alignment Engine

Connect projects to actual job requirements and turn implementation into credible evidence.

- Role-specific project briefs
- Required technical competencies
- Implementation checkpoints
- Project evidence
- Resume bullet suggestions based on verified work

05 Interview Readiness Engine

Evaluate whether someone can demonstrate their preparation clearly in an interview.

- Technical question assessments
- Project explanation assessments
- Behavioural preparation
- Weak-area identification
- Readiness reports

The connection is the product

The important part is that these are not five disconnected features. They should share information. For example, a missing database skill identified by the Job Intelligence Engine should be available to the Preparation Engine as a learning task, to the Skill Verification Engine as an assessment, to the Project-to-Job Alignment Engine as a project requirement, and to the Interview Readiness Engine as a topic to explain.

JOB INTELLIGENCE	PREPARATION	VERIFICATION	PROJECT ALIGNMENT	INTERVIEW
Find the gap	Learn and practise	Test ability	Build proof	Demonstrate it

Information should move both ways: assessment results can change the plan; project evidence can strengthen a skill record; interview results can reveal a new preparation gap.

PRODUCT CHANGE 02

Make My Plan job-oriented

The three entry points already identified are good and should be retained. They let different learners start from the context that matters to them.

ENTRY POINT	USER INTENT
From Role Analyzer	I have a target job or job description. Help me close the gaps.
Placement preparation	I am preparing for placements and need a structured route.
Switching to a target role	I want to move into a different role and need to build the required skills.

Tasks should have a purpose and an outcome

A task should describe what the learner will understand, practise or produce—not merely which page to open and how long to stay there.

LESS USEFUL	MORE USEFUL
Study DBMS indexing — 45 minutes	**Understand B-tree indexing, solve two SQL/indexing exercises, and explain when an index hurts write performance.**

Completion should mean more than opening a page and ticking a checkbox. A learner may complete the reading task but still need to solve the exercises or explain the concept. Keep task completion separate from demonstrated skill.

A useful job-oriented task can contain

- **Purpose:** which job requirement or competency this task supports.
- **Action:** what the learner must read, solve, build or explain.
- **Outcome:** what they should be able to do afterwards.
- **Evidence:** how PrepStack will record or check that outcome.
- **Next step:** what happens if the learner succeeds or needs more practice.

This makes My Plan an execution layer for job preparation—not simply a calendar or generic to-do list.

ARCHITECTURE FOUNDATION

Make the Skill Catalog the central intelligence layer

PrepStack's existing 120-skill catalog already maps skills to internal resources. It could become one of the most important parts of the architecture: the shared vocabulary that allows every system to refer to the same skill in a consistent way.

Instead of each module keeping its own separate list of skills, every module should reference the canonical Skill Catalog.

JOB INTELLIGENCE	PREPARATION	SKILL VERIFICATION
Role requirements Skill gaps	Notes Roadmaps My Plan	Assessments Evidence
PROJECT ALIGNMENT	**SHARED SKILL CATALOG**	**INTERVIEW READINESS**
Project briefs Artifacts	One canonical definition per skill	Questions Readiness reports

What each skill could eventually contain

Prerequisites What should be understood first?	**Related skills** Which other competencies connect to this one?
Learning resources Which notes, roadmap steps or external resources teach it?	**Practical exercises** How can the learner apply it?
Assessment criteria What does a satisfactory demonstration look like?	**Project applications** Where can this skill be used in a real project?
Interview questions How might an interviewer test or discuss it?	**Evidence requirements** What proof supports the learner's skill status?

This lets multiple modules use the same skill definitions. A skill such as **DBMS indexing** can be referenced by a Role Analyzer gap, a DBMS lesson, a My Plan task, a practical exercise, a project checkpoint and an interview question—without duplicating the skill definition.

> **Architecture principle: the Skill Catalog defines what a skill is; the other systems define how that skill is discovered, learned, practised, demonstrated and communicated.**

PRODUCT CONNECTION

Connect AI project generation with the Role Analyzer

The current project generator accepts a technology stack, domain and complexity. That is a useful starting point. The next evolution is to let the learner generate a project around the requirements of a target job.

NEW GENERATION PATH

Generate a project based on my target job

Use the Role Analyzer's missing or underdeveloped skills as project requirements. Generate a project brief designed to help the learner practise and demonstrate those competencies.

Example workflow

01 The learner uploads a resume and pastes a Backend Developer job description.

02 The Role Analyzer identifies required competencies and highlights missing skills.

03 The learner chooses "Generate a project based on my target job".

04 The generator selects a manageable subset of relevant missing skills.

05 It creates a project brief with technical requirements and implementation checkpoints.

06 The learner builds the project and records repository, deployment and checkpoint evidence.

07 PrepStack maps the completed work back to the relevant skills and helps draft resume bullets based on what was actually implemented.

What makes the project brief useful

- A real problem and target user.
- The selected job competencies the project is intended to demonstrate.
- Functional requirements and clear acceptance criteria.
- Implementation milestones and review checkpoints.
- Technical trade-offs the learner should understand and explain.
- A final evidence checklist: repository, README, demo, tests and explanation.
- Resume bullet suggestions grounded in completed work—not invented metrics or unverified claims.

The aim is not to generate a random impressive-sounding project. It is to help the learner build something that fills a real skill gap and produces evidence they can discuss.

IMPLEMENTATION STRATEGY

Where to prioritise investment

Do not try to implement all five systems immediately. Build the foundations first, connect the preparation workflow second, and add verification and readiness capabilities only when the underlying content and skill mappings are reliable.

PHASE 1

Make existing foundations excellent

- Finish CS notes and roadmaps.
- Improve resume and behavioural guides.
- Build shared progress records.
- Persist problem notes.
- Improve the Role Analyzer mappings.

PHASE 2

Connect preparation

- Shared checklist.
- My Plan.
- Job-specific task generation.
- Skill Catalog improvements.
- Meaningful dashboard analytics.

PHASE 3

Build differentiation

- Skill verification.
- Practical competency assessments.
- Project-to-job alignment.
- Evidence-based skill reports.

PHASE 4

Interview readiness

- Project explanation assessments.
- Technical interview simulations.
- Behavioural answer evaluation.
- Readiness history and improvement tracking.

How to know each phase is ready

PHASE	EXIT CHECK
1	Core content is accurate, useful and mapped to the Skill Catalog; progress and notes are reliable.
2	A target role or preparation goal can produce a coherent plan with tasks linked to actual skills and outcomes.
3	Important skills can be assessed, practical work can be linked to competencies, and evidence can be reviewed.
4	Learners can practise explaining technical knowledge, projects and behavioural examples, then see specific weak areas.

THE NORTH STAR

One connected journey. Not five separate features.

The five systems become powerful when they share a common understanding of the learner's target role, required skills, preparation activity and evidence.

01

Understand the job

What does the target role actually require?

02

Prepare with purpose

Which learning and practice actions close the gaps?

03

Verify the skill

Can the learner explain and apply the concept?

04

Build proof

Can the learner demonstrate it in a real project?

05

Show readiness

Can the learner communicate that ability in an interview?

The Skill Catalog is the shared intelligence layer. My Plan turns skill gaps into purposeful work. Verification checks whether learning has become ability. Project alignment turns ability into practical evidence. Interview readiness helps the learner communicate that evidence.

PrepStack's product promise: help learners understand what their target job requires, prepare for those requirements, build proof of their skills, and demonstrate their readiness with confidence.

A focused product architecture for the next stage of PrepStack.